

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELGAUM-590014**



**A Mini Project for Creative Assessment Report on
“MURESY (Music Recommendation System)”**

*Submitted in partial fulfillment of the requirements for the award of the degree of
Bachelor of Engineering in Artificial Intelligence and Machine Learning*

carried out

At

Department of Artificial Intelligence and Machine Learning, DSATM

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**DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND
MANAGEMENT**

**Department of Artificial Intelligence and Machine Learning
(Accredited by NACC A+ Grade)**

Udayapura, Kanakapura Road, Bangalore-560082
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CERTIFICATE

Certified that the Mini Project work entitled “**MURESY (Music Recommendation System)**” carried out by **AAHISH AAYAN(1DT21AI001)**, **ABHIJEET SAURABH(1DT21AI002)**, **HARDIK KUMAR(1DT21AI023)** & **OORJA SAXENA(1DT21AI043)** are the students of **Dayananda Sagar Academy of Technology and Management** in partial fulfilment for the award of Bachelor of Engineering in **Artificial Intelligence and Machine Learning** of the Visvesvaraya Technological University, Belgaum during the year 2022-23. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The Mini Project report has been approved as it satisfies the academic requirements in respect of Internship work prescribed for the said Degree.

Signature of Coordinator

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STUDENT DECLARATION

We, Students **AAHISH AAYAN(1DT21AI001), ABHIJEET SAURABH(1DT21AI002), HARDIK KUMAR(1DT21AI023) & OORJA SAXENA(1DT21AI043)** hereby declare that the presented report of workshop titled “**MURESY (Music Recommendation System)**” is uniquely prepared by us after the completion of “Three days workshop on Machine Learning” carried out at “**DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT**”.

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CHAPTER 1

INTRODUCTION

This Mini project is a content-based music recommendation based on user-specific classification problem. This model uses Machine Learning techniques to recommend music according to the user's previous data. The model will then filter out the same music and recommends music according to the user's mood.

Pre-Requisites :-

1. Python:

Python is the most widely used programming language for machine learning.

2. Integrated Development Environment (IDE):

Jupyter Notebook: Ideal for interactive coding and data exploration.

Anaconda: A powerful IDE designed specifically for Python development.

3. Libraries and Frameworks:

Install the following libraries and frameworks using Python's package manager, pip:

- NumPy: For numerical computations and array operations.
- Pandas: For data manipulation and analysis.
- Scikit-learn: For machine learning algorithms and model evaluation.
- Matplotlib and Seaborn: For data visualization.
- Cosine Similarity: For finding the similar music
- Count Vectorizer: For counting the data

CHAPTER 2

ROLE OF THE STUDENT

In this project, Python is used for all the features of the **Music Recommendation System**. Each individual student gained the knowledge of Machine Learning. Mainly, four Python libraries are used to build the Music Recommendation System namely Numpy, Pandas, Matplotlib, and Seaborn.

- The basic introduction to Python.
- Basic introduction for creating ML model.
- Knowledge about different python libraries and data frameworks.
- Basic knowledge of Linear Regression Models.
- Basic Knowledge of Jupyter Notebook.
- Basic Knowledge about Anaconda.
- Basic knowledge about numpy, pandas, matplotlib and seaborn.
- Using different libraries for MURESYS.

CHAPTER 3

SCHEDULE OF WORKSHOP

Date: 24/08/2023 to 26/08/2023

Time: 9.30 am to 5.00pm

- **DAY-1(24/08/2023)** :- Introduction to Machine Learning, basics of numpy and pandas. Introduction to Anaconda Environment and jupyter notebook.
- **DAY-2(25/08/2023)** :- Data Exploration basics, One complete data exploration project exercise and One assignment.
- **DAY-3(26/08/2023)** :- Linear Regression Model, Single and Multiple linear regression and other basic models.

CHAPTER 4

TOOLS AND TECHNOLOGY USED

4.1 PYTHON

Python is considered a high-level programming language, which means it abstracts many low-level details like memory management, allowing developers to focus on solving problems rather than dealing with low-level intricacies. Python is an interpreted language, which means you don't need to compile your code before running it. You write Python code in plain text, and the Python interpreter executes it directly. This feature makes development and debugging more straightforward.

4.2 NUMPY

NumPy is a fundamental Python library for numerical and scientific computing. It provides support for arrays (including multi-dimensional arrays) and matrices, as well as a wide range of mathematical functions to operate on these arrays efficiently. NumPy is an essential tool for data scientists, engineers, and researchers working on numerical and scientific computing tasks in Python.

4.3 PANDAS

"Pandas" is a Python library widely used for data manipulation and analysis. It provides data structures and functions for working with structured data, such as tabular data, time series, and more. Pandas is a fundamental tool in the data science and data analysis toolkit because it makes it easier to clean, explore, and analyze data efficiently. Pandas is an essential tool for data wrangling and analysis tasks in Python. It simplifies many common data-related operations and provides a convenient interface for exploring and transforming data, making it an invaluable resource for data scientists, analysts, and anyone working with data in Python.

4.4 MATPLOTLIB

Matplotlib is a widely used Python library for creating static, animated, and interactive data visualizations in 2D and, with some extensions, 3D. It is one of the primary tools for data visualization in Python and is particularly popular among data scientists, analysts, and researchers for creating plots, charts, and graphs to convey information and insights from data. Matplotlib provides a versatile and powerful way to visualize data, making it easier to explore and communicate patterns and trends in your data.

4.5 SEABORN

Seaborn is a Python data visualization library built on top of Matplotlib. It is designed to create attractive and informative statistical graphics, making it easier to visualize and understand complex datasets. Seaborn simplifies the process of creating various types of statistical plots and offers a high-level interface for creating aesthetically pleasing visualizations with minimal coding effort. Seaborn is a valuable tool for anyone working with data analysis and data visualization in Python. It offers an elegant and efficient way to create informative, publication-quality visualizations with a focus on statistical insights. Additionally, Seaborn works well with other data analysis libraries like Pandas and NumPy, making it a popular choice for data scientists and analysts.

4.6 ANACONDA

Anaconda Navigator, a graphical user interface, allows users to manage packages, environments, and Jupyter Notebooks. Users can also choose to work with various Python IDEs such as Spyder, which is included in the Anaconda distribution. Anaconda is a popular choice among data scientists, analysts, and researchers because it provides a convenient way to set up a Python environment for data analysis and machine learning. It simplifies the often complex process of managing dependencies and libraries, making it easier to get started with data-related projects.

4.7 JUPYTER NOTEBOOK

Jupyter Notebook is an interactive web-based environment for creating and sharing documents that contain live code, equations, visualizations, and narrative text. Jupyter Notebook is widely used for data exploration, analysis, and sharing research findings.

CHAPTER 5

REFLECTIONS

This was the project which was completed under the domain of Machine Learning using Python. The objective of this project was to build a Music Recommendation System which recommends music according to the user’s mood.

OUTCOMES:

	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	valence	...
0	0.831	0.814	2	-7.364	1	0.4200	0.0598	0.013400	0.0556	0.3890	... 2Vc6NJ9PW9gD9q34
1	0.719	0.493	8	-7.230	1	0.0794	0.4010	0.000000	0.1180	0.1240	... 7pgJBLVz5VmnL7uG
2	0.850	0.893	5	-4.783	1	0.0623	0.0138	0.000004	0.3720	0.0391	... 0vSWgAlfpye0WCGeN
3	0.476	0.781	0	-4.710	1	0.1030	0.0237	0.000000	0.1140	0.1750	... 0VSNJqQkwuH2ei1r
4	0.798	0.624	2	-7.668	1	0.2930	0.2170	0.000000	0.1660	0.5910	... 4jCeguq9rMTIbMmPI

5 rows × 22 columns

fig 5.1 CSV file

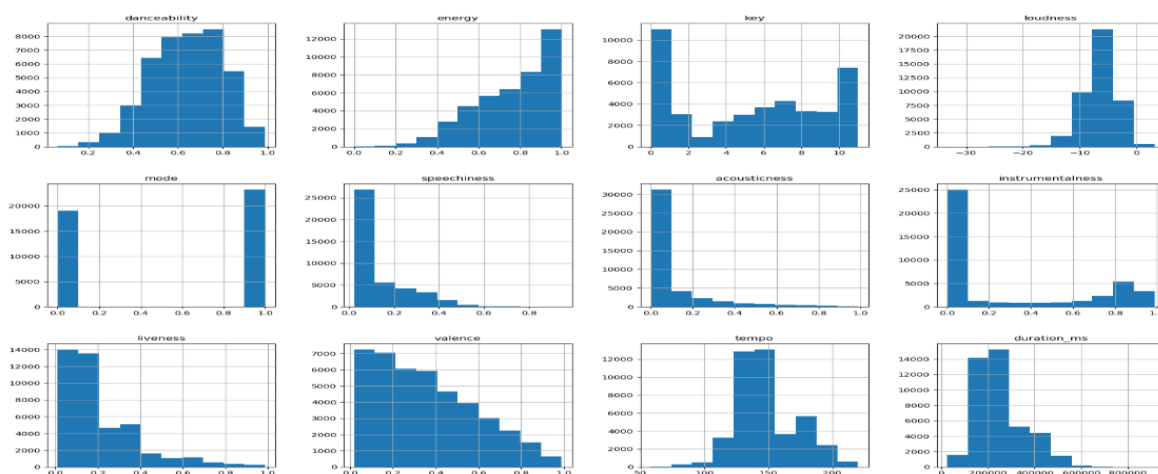


fig 5.2 Frequency of different song types

Workshop Report on “MURESY”

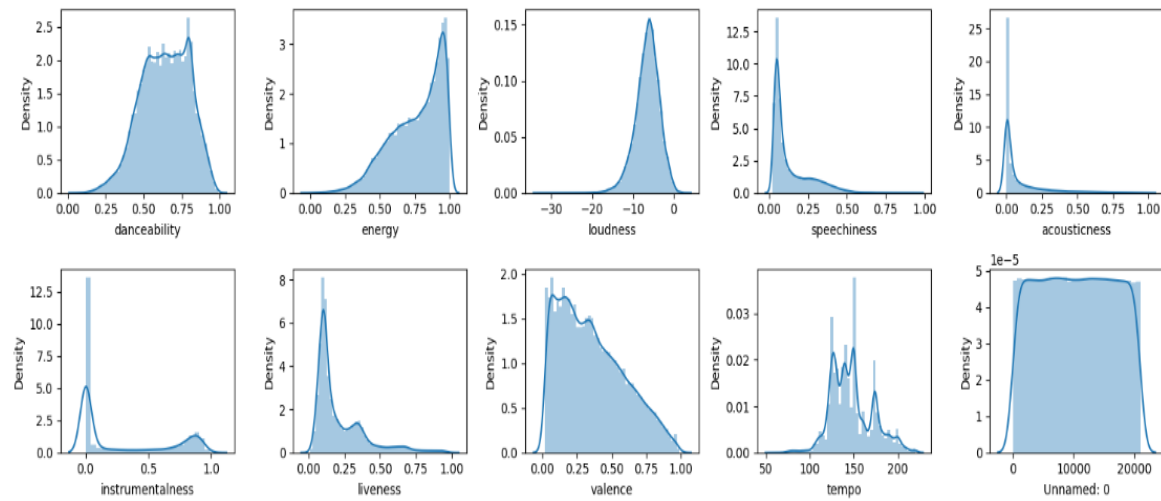


fig 5.3 Similar Songs

```
[13]: recommend_songs('Closer_RnB')
```

This song is either not so popular or you\have entered invalid_name.
Some songs you may like:

nan
The American Dream
nan
Chop Suey!
nan

```
[14]: recommend_songs('The_Real_Slim_Shady')
```

This song is either not so popular or you\have entered invalid_name.
Some songs you may like:

nan
Call Me Hopeless, but Not Romantic
When the Body Goes Cold
nan
FlatBed Freestyle

fig 5.4 Output

WORKING CODE

```
import numpy as np
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sb
```

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
from sklearn.feature_extraction.text import CountVectorizer
```

```
from sklearn.manifold import TSNE
```

```
import warnings
```

```
warnings.filterwarnings('ignore')
```

```
In [2]:
```

```
tracks = pd.read_csv('songs.csv')
```

```
tracks.head()
```

```
tracks.shape
```

```
tracks.info()
```

```
#model = TSNE(n_components = 2, random_state = 0)
```

```
#a = pd.read_csv('songs.csv')
```

```
#tsne_data = model.fit_transform(a.head(500))
```

```
#plt.figure(figsize = (7, 7))
```

```
#plt.scatter(tsne_data[:,0], tsne_data[:,1])
```

```
#plt.show()
```

In [6]:

```
tracks['song_name'].nunique(), tracks.shape
```

```
tracks.hist(figsize =(20,20))
```

```
plt.title("Histogram plot", size =15, weight = 'bold')
```

```
plt.subplots(figsize = (15, 5))
```

```
for i, col in enumerate(floats):
```

```
    plt.subplot(2, 5, i + 1)
```

```
sb.distplot(tracks[col])
```

```
plt.tight_layout()
```

```
plt.show()
```

```
def get_similarities(song_name, data):
```

```
# Getting vector for the input song.
```

```
text_array1 =
```

```
song_vectorizer.transform(data[data['song_name']==song_name]['genre']).toarr
```

```
ay()
```

```
num_array1 =
```

```
data[data['song_name']==song_name].select_dtypes(include=np.number).to_n
```

```
umpy()
```

```
# We will store similarity for each row of the dataset.
```

```
sim = []
```

```
for idx, row in data.iterrows():
```

```
    name = row['song_name']
```

```
    # Getting vector for current song.
```

```
    text_array2 =
```

```
    song_vectorizer.transform(data[data['song_name']==name]['genre']).toarray()
```

```
    num_array2 =
```

```
    data[data['song_name']==name].select_dtypes(include=np.number).to_numpy(
    )
```

```
    # Calculating similarities for text as well as numeric features
```

```
    text_sim = cosine_similarity(text_array1, text_array2)[0][0]
```

```
    num_sim = cosine_similarity(num_array1, num_array2)[0][0]
```

```
    sim.append(text_sim + num_sim)
```

return sim

In [11]:

```
def recommend_songs(song_name, data=tracks):
```

```
    # Base case
```

```
    if tracks[tracks['song_name'] == song_name].shape[0] == 0:
```

```
        print("This song is either not so popular or you\have entered invalid_name.\n
```

```
Some songs you may like:\n')
```

```
    for song in data.sample(n=5)['song_name'].values:
```

```
        print(song)
```

```
    return
```

```
data['similarity_factor'] = get_similarities(song_name, data)
```

```
data.sort_values(by=['similarity_factor','genre'],
```

```
                ascending = [False, False],
```


First song will be the input song itself as the similarity will be highest.

```
display(data[['song_name']][2:7])
```

In [12]:

```
recommend_songs('Uptown Funk (ft.Bruno Mars)')
```

```
recommend_songs('Closer_RnB')
```

```
recommend_songs('The_Real_Slim_Shady')
```

```
recommend_songs('Gucci_Gang')
```

```
recommend_songs('Remember You (feat. The Weeknd)')
```

CONCLUSION

With this hands on 3 day workshop we got to learn about Machine Learning, Python and it's various libraries and data frame works such as numpy, pandas, matplotlib and seaborn and how to use it in descriptive analysis. With the help of all this information we successfully built a music recommendation system.

In conclusion, the three-day workshop in machine learning has been an incredibly enriching and transformative experience for all participants. Over the course of these three days, we delved deep into the fascinating world of machine learning, exploring its fundamental concepts, algorithms, and practical application.

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